# Textual Analysis in Finance

*... or in whatever you like*

## Week 1: Overview of Textual Analysis in Finance

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# Welcome

# Course overview

- This course introduces **textual analysis** (also known as *Natural Language Processing*, NLP) and its **applications in finance**:
  - focuses on extracting economically meaningful information from financial texts, such as sentiment and topic classification
  - other NLP tasks (e.g., text segmentation, named entity recognition, question answering) may be mentioned, but are not the primary focus

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  - identify economically relevant information in textual data
  - understand core textual analysis techniques
  - use text data to answer specific financial questions

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  - identify economically relevant information in textual data
  - understand core textual analysis techniques
  - use text data to answer specific financial questions
- Expectations:
  - be ready to get your hands dirty
  - review materials before each lecture, especially the **Jupyter notebooks**
  - actively ask questions

# Course organization

- Course format:
  - 6 weeks, 4 hours per week
  - First half (of each lecture): theory and concepts
  - Second half: hands-on coding and applications

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- Course agenda:

Week 1 Course overview, Bag-of-Words
Week 2 Sentiment analysis and topic modeling
Week 3 Neural embedding
Week 4 Language models
Week 5 Large language models
Week 6 Retrieval-augmented generation, feedback, guest lecture

# Miscellaneous

- Contact: `triminh.phan@unibas.ch`
- Course materials:
  - Syllabus: Papers cited in course slides (you don't need to read all)
  - Slides and code: released on ADAM **at least 2 days** before each lecture
  - Github page: materials + supportive code for the coding sessions
- Assessment:
  - Coding assignment (individual): 40%
  - Final project (groups of up to 3): 60% (same grade)

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- Important skills: Strings, lists, dictionaries, arrays, loops, functions, objects
- Relevant libraries:
  - `numpy`, `pandas`, `scipy` - Basic data analytics
  - `nltk`, `spaCy`, `gensm` - Text analysis
  - `scikit-learn`, `xgboost` - Machine learning
  - `huggingface` - Pre-trained language models
  - `transformers`, `pytorch` - Deep learning

# Text as data

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- **Immediate market impact**: Words in earnings calls, press releases, or central bank statements can move prices faster than official numbers
- **Abundance**: various data sources (10-Ks, earnings calls, news, policy statements) with thousands/millions of documents

# Example - FED speech

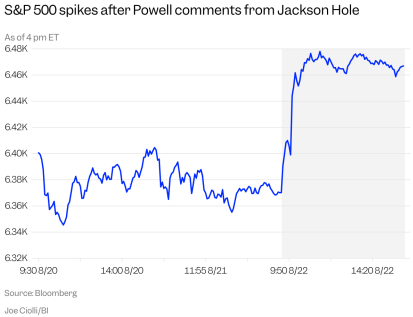


Figure: S&P 500 index during FED chair, Jerome Powell, in Jackson Hole, on August 22, 2025. Source: Bloomberg.

*... The stability of the unemployment rate and other labour market measures allows us to proceed carefully as we* ***consider changes to our policy stance*** *...*

# Example - Tesla Q1 2025 earnings call transcripts

Figure: Tesla stock reaction following a disappointing Q1 2025 earnings report. Source: Reddit.

Elon Musk: *... at Tesla, we've gone through many near-death experiences ... if we do execute well, I think Tesla will be the* ***most valuable company*** *in the world by far. It may be* ***as valuable as the next five companies combined*** *... but* ***there'll be a few bumps*** *along the road before that happens ... then once it does start to move the financial needle in a significant way, it will really* ***go exponential*** *from there ...*

# Why not text data?

- Text data is NOISY!!!!!
  - Typos, e.g., rec**ie**vables vs rec**ei**vables
  - Informal talks, e.g., "**Well**, there's never a dull moment these days" (Elon Musk, 2025 Q1 earnings call)
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- NLP is (very) costly

# Basic assumptions

Doing textual analysis involves implicitly making the following assumptions:

- **Texts contain relevant and meaningful information**, for example:
  - Li et al. [2021] quantify ***corporate culture*** from earnings call transcripts
  - Florackis et al. [2023] quantify ***cybersecurity risk*** from 10-K/Q filings
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- **Texts can be represented by features**, such as words, tokens, or embeddings
  - This is where some models fail, and more advanced models are developed to overcome these limitations

# From Raw Text to Features

# Textual Analysis Pipeline

- **Raw text** - Raw data
- **Normalization** - Clean and standardize the raw text
- **Features** - Turn clean text into numbers
- **Model** - Learn patterns from the text-based numerical data
- **Output** - Predictions or insights

# Raw text

- **Corporate disclosures (10-K/Q filings)** / Example
  - Characteristics: standardized, formal, lengthy, repetitive, legally constrained
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- **Central bank communication** / Example [Meme]
  - Characteristics: spoken but carefully crafted, institutional, low-frequency
  - Content: policy stance, macroeconomic outlook
  - Data access: FED's Greenbook, FOMC minutes, ECB, BIS

# Text normalization

A process of **cleaning** and **standardizing** raw text. Basic steps include:

- **Case normalization** (lowercasing) - Reduce the vocabulary size
  "**I**ncome increased by 12.5% in **Q**4" → "**i**ncome increased by 12.5% in **q**4"
- **Punctuation and number removal** - Reduce noise
  "income increased by **12.5%** in q**4**" → "income increased by in q"
- **Stopword removal** - Reduce generic words
  "income increased **by in** q" → "income increased q"
- **Lemmatization or stemming** - Normalize word forms
  "income increase**d** q" → "income increase q" (lemmatization)
  "incom**e** increas**ed** q" → "incom increas q" (stemming)
- **Tokenization** - Define units "income increase q" → ["income", "increase", "q"]

# Text normalization in finance - Caveats

- **Normalization must be done with care–especially in finance [Loughran and McDonald, 2016]**
  - Lowercasing may destroy meaning, e.g., "US" → "us"
  - Numbers often carry economically meaningful information
  - Punctuation (e.g., periods, commas) helps define sentence structure
  - Stopwords may be informative:
    - "will" and "about to" indicate forward-looking statements
    - "may", "might", and "would" reflect uncertainty
  - Stemming is often too aggressive for financial text; lemmatization is preferred
  - Additional cleaning steps may be required depending on the data, e.g., removing extra whitespaces; cleaning HTML tags or non-Unicode characters

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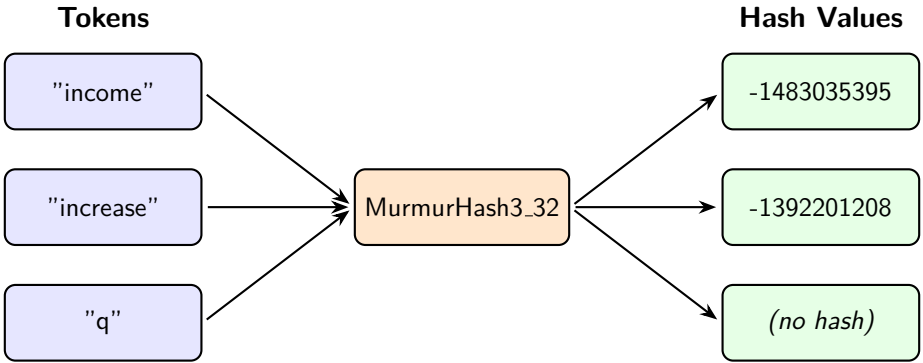
- **General advice**
  - Always inspect the raw text before normalization
  - Use domain-specific normalization rules
  - Avoid heavy normalization when using language models

# Hashing

How to convert "**income increase q**" into numbers?

**Idea:** Convert each word (token) into a unique number by a **hash function**



# Hashing

| Input sentences | | Hashed values |
|---|---|---|
| s1: "income increase q" | *Hashing* → | [-1483035395, -1392201208] |
| s2: "operate expense decrease significantly" | *Hashing* → | [732497022, -1121816754, -322192674, 1119211873] |

- **Not ready** for modeling due to **misaligned dimensions**!

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- **Not ready** for modeling due to **misaligned dimensions**!
- Choose a dimensionality $D$ and map hashed values using % $D \rightarrow$ **hashing trick**!

# Hashing trick

Choose $D = 8$, initially, every sentence can be represented by a vector of 8 zeros

$$[0, 0, 0, 0, 0, 0, 0, 0]$$

$f(\text{“income”}) = -1483035395 \rightarrow \text{its index} = |-1483035395| \ \% \ 8 = 3$

Subtract 1 from the $4^{th}$ index (because the hashed value of “income” is negative), and the vector becomes

$$[0, 0, 0, -1, 0, 0, 0, 0]$$

Repeat for all words and all sentences!

# Hashing - Pros & Cons

The two sentences after the hashing process (and $L_2$ normalization) become

$$\begin{bmatrix} \text{s1} \\ \text{s2} \end{bmatrix} = \begin{bmatrix} -0.7071 & 0 & 0 & -0.7071 & 0 & 0 & 0 & 0 \\ 0 & 0.4082 & -0.8165 & 0 & 0 & 0 & 0.4082 & 0 \end{bmatrix}$$

- Pros:
  - Computationally efficient
  - Fixed dimensions - $D$ is up to our choice
  - Robust to new words - everything can be hashed

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- Pros:
  - Computationally efficient
  - Fixed dimensions - $D$ is up to our choice
  - Robust to new words - everything can be hashed
- Cons:
  - No interpretability - rarely used in economics/finance!
    - Dimensions have no meaning
    - Can't convert a vector back into human-readable text
  - Collisions - Two different words can share the same index

# Bag-of-Words

## What is the Bag-of-Words (BoW) model?

Unlike the Hashing, Bag-of-Words (BoW) relies on a **predefined vocabulary** (dictionary/lexicon). Each word is mapped to a specific, permanent index based on its position in the vocabulary.

- (+) ***Interpretable*** (Index = Word)
- (-) Need to build a vocabulary beforehand
- (-) Size = Vocabulary length

# Bag-of-Words

Given two sentences: **"income increase q"**, **"operate expense decrease significantly"**

## Step 1: Build the vocabulary

$V = [\text{"decrease"}, \text{"expense"}, \text{"income"}, \text{"increase"}, \text{"operate"}, \text{"q"}, \text{"significantly"}]$

## Step 2: Build a so-called **document-term** matrix

$$\begin{array}{c} \\ s1 \\ s2 \end{array} \begin{array}{c} \begin{array}{ccccccc} \text{decrease} & \text{expense} & \text{income} & \text{increase} & \text{operate} & \text{q} & \text{significantly} \end{array} \\ \begin{bmatrix} 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

Dimensions: $N \times |V|$, where $N$ is the number of "documents" and $|V|$ is the number of "terms" (vocab size)

# Bag-of-Words - Pros & Cons

- Pros:
  - **High interpretability** (big plus)
  - Easy, simple, and efficient

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- Pros:
  - **High interpretability** (big plus)
  - Easy, simple, and efficient
- Cons:
  - More frequent words dominate (more important) - suffers from **Zipf's law** [Manning and Schutze, 1999]
  - Ignores word order, semantics, and contexts
    - *sales increase, costs decrease* = *sales decrease, costs increase*
    - *sales increased significantly* ≠ *sales rose sharply*
    - *the bank by the river* vs. *financial bank*
  - Vocabulary-dependent: unseen words cause out-of-vocabulary errors
  - High dimensionality: large and sparse document-term matrix (vocabulary can grow to millions)

# Term frequency - Inverse document frequency (TF-IDF)

Bag-of-Words suffers from Zipf's law $\rightarrow$ more frequent words take higher weights
TF-IDF:

$$tf.idf(w,d) = tf(w,d) \times idf(w) = \frac{BoW(w,d)}{l_d} \times \log\frac{N}{df(w)},$$

where:

- $BoW(w,d)$: the frequency of word $w$ in document $d$
- $l_d$: the length of document $d$
- $df(w)$: the number of documents containing $w$
- $N$: the total number of documents

# TF-IDF

- BoW counts words, but ignores how informative they are
- TF-IDF reweights words by their importance across documents
- Other issues of BoW remain unsolved!

# Coming next

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- Hands-on practice
  - Normalizing NVDA earnings call transcripts
  - Extract features using `HashVectorizer`, `CountVectorizer`, and `TfidfVectorizer`
  - Visualization - Word cloud
  - Notebook
- Next lecture
  - Sentiment analysis
  - Topic models

# References I

Maximilian Ahrens, Deniz Erdemlioglu, Michael McMahon, Christopher J Neely, and Xiye Yang. Mind your language: Market responses to central bank speeches. *Journal of Econometrics*, 249:105921, 2025.

Chris Florackis, Christodoulos Louca, Roni Michaely, and Michael Weber. Cybersecurity risk. *The Review of Financial Studies*, 36(1):351–407, 2023.

Kai Li, Feng Mai, Rui Shen, and Xinyan Yan. Measuring corporate culture using machine learning. *The Review of Financial Studies*, 34(7): 3265–3315, 2021.

Tim Loughran and Bill McDonald. Textual analysis in accounting and finance: A survey. *Journal of Accounting Research*, 54(4):1187–1230, 2016.

Christopher Manning and Hinrich Schutze. *Foundations of statistical natural language processing*. MIT press, 1999.